

Name: Tiger-AI Effectiveness Assessment & Monitoring System (TEAMS)

Goal: Developing an Automated Effectiveness Metric for Project Tiger Conservation Efforts

1. Executive Summary

Project Tiger, initiated in 1973, in India, has grown from 9 to 58 reserves as of 2025, with the goal of maintaining viable tiger populations and balancing human-wildlife coexistence. While the project has achieved a population increase to 3,682 tigers in 2022, challenges such as poaching, deforestation, and administrative irregularities persist.

This technical deliverable outlines the requirements for TEAMS, an AI-driven system designed to ingest historical census data, camera trap imagery, and GIS logs to mathematically score the effectiveness of specific tiger reserves and the overall project.

2. Problem Statement

Current assessment relies on periodic four-year census cycles and manual "combing" of forest floors. There is a need for a continuous, data-driven "Effectiveness Index" that correlates administrative interventions (reserve expansion, relocation) with biological outcomes (population density, prey base) to objectively measure success.

3. Data Sources & Integration

The system must ingest and normalize data from the following authoritative sources:

- Historical Census Data: Population estimates ranging from 1,141 (2006) to 3,682 (2022).
- Geospatial Data (GIS): Shapefiles of 58 designated tiger reserves, specifically distinguishing between Core Areas (protected/national parks) and Buffer Zones (mixed forest/non-forest land).
- Surveillance Feeds: Data from existing camera traps, infrared thermal cameras, and the "MSTrIPES" (Monitoring System for Tigers – Intensive Protection & Ecological Status) program.
- Threat Logs: Records of poaching incidents, human-wildlife conflicts, and deforestation alerts.

4. Technical Modules & Deliverables

Module A: Population Viability Analysis (Computer Vision)

- Objective: Automate the processing of camera trap data to validate census accuracy.
- Task: Train a Convolutional Neural Network (CNN) to identify tigers, co-predators, and prey animals from trap images.
- Effectiveness Metric: Calculate the Prey-Predator Ratio per reserve. According to Section 38 of the Wildlife Protection Act, reserves must maintain a viable population of tigers alongside co-predators and prey.
- Success Indicator: The model should detect population dips in specific reserves before the official 4-year census.
- Data source: The model should detect population dips in specific reserves before the official 4-year census.

Module B: Habitat Integrity Monitor (Remote Sensing/GIS)

- Objective: Measure the effectiveness of "Core vs. Buffer" management strategies.
- Task: Implement satellite imagery segmentation to detect unauthorized construction, deforestation, or infrastructure encroachment (e.g., highways) in defined buffer zones.
- Effectiveness Metric: Habitat Fragmentation Score. A declining score indicates a failure to protect the habitat connectivity required for the tiger's survival.

Module C: Anti-Poaching Resource Optimization (Predictive Modeling)

- Objective: Assess if the deployment of the Special Tiger Protection Force is effective.
- Task: Correlate patrol paths (from MSTRIPES data) with poaching incident locations. Use clustering algorithms to identify "unpatrolled high-risk zones."
- Effectiveness Metric: Intervention Latency. Measure the time reduction between potential threat detection (via thermal/wireless sensors) and staff deployment.

Module D: The "Tiger Success Index" (Statistical Regression)

- Objective: Create a single composite score for Project Tiger's overall health.
- Task: Build a regression model that weighs the following variables:
 - Input: Reserve Area Expansion (from 9,115 km² in the 1980s to current coverage).
 - Input: Funding/Grant allocation.
 - Output: Net population growth (currently ~30% increase observed between 2010-2014).
- Deliverable: A dashboard showing which specific reserves are underperforming relative to their size and funding.

5. Success Criteria for the Evaluation

The project will be considered successful if you have

1. Accurately replicate the 2006-2022 population growth trend (from 1,141 to 3,682) using only raw input data.
2. Successfully completed Module D and at least one other module amongst A, B and C.
3. Bonus: Identify "at-risk" reserves where human-animal conflict in buffer zones is statistically likely to increase based on current forest rights and dwelling data.